

Subject Description Form

Subject Code	EE528																								
Subject Title	System Modelling and Optimal Control																								
Credit Value	3																								
Level	5																								
Pre-requisite/ Co-requisite/ Exclusion	Nil																								
Objectives	- To provide students with a sound knowledge of system identification and modelling techniques in areas of prediction and control. - To introduce modern control design techniques.																								
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Model systems using State Variable and Transfer Functions. - Design and optimize optimal controllers for system models. - Predict and control techniques for model systems. - Apply control system in practical applications.																								
Subject Synopsis/ Indicative Syllabus	System models: functions, transformations and mapping, Laplace transformation and z-transformation, state variables and state space models of dynamic systems, relations between state space models and transfer function models, solutions of unforced linear state equations, matrix exponential, eigenvalues and eigenvectors, Jordan form, solutions of linear state equations, transition matrix. Stability, controllability, and observability: stability, Lyapunov stability, Lyapunov function, controllability and observability, definition and criteria, stabilizability and detectability, feedback control. Optimal control: Calculus of variations, formulation of optimal control problems, Pontryagin maximum principle, Riccati equation, application to linear regulator.																								
Teaching/Learning Methodology	Basic concepts and theories are taught in lectures and tutorials. The students are expected to solve theoretical and practical control problems with critical and analytical thinking. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>Lectures</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>Tutorials</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>Assignments</td><td></td><td></td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> </tbody> </table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lectures	✓	✓	✓		Tutorials	✓	✓	✓		Assignments			✓	✓
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Tutorials	✓	✓	✓																						
Assignments			✓	✓																					

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Examination	60%	✓	✓	✓	
	2. Assignments	40%	✓	✓	✓	✓
	Total	100%				
	The outcomes on concepts, analytical skills, problem-solving techniques, design and applications, and practical considerations of designing control systems are assessed by the usual means of examination and assignments.					
Student Study Effort Expected	Class contact:					
	▪ Lecture/Tutorial				39 Hrs.	
	Other student study effort:					
	▪ Reading and studying				43 Hrs.	
	▪ Completing assignments				23 Hrs.	
	Total student study effort				105 Hrs.	
Reading List and References	1. L. Ljung, System Identification: Theory for the User (2nd Edition), Prentice Hall. 2. Karl J. Åström, Björn Wittenmark, Adaptive Control, Dover Publications. 3. N. Nise, Control Systems Engineering (7th Edition), John Wiley & Sons. 4. Gordon Blower, Linear Systems, Springer Cham.					